

Flexible-margin Gaussian-copula full random-effects models: likelihood-based estimation by stochastic approximation

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Abstract

Full random-effects modelling (FREM) jointly describes individual parameters and covariates using a multivariate Normal population distribution. We extend FREM by combining Gaussian dependence with prespecified continuous marginal distributions for parameter random effects and continuous or ordered categorical marginal distributions for covariates. The model reduces to Gaussian FREM when all margins are Normal and permits missing covariates. We derive the observed-data likelihood and estimate population locations, marginal-distribution parameters, dependence parameters and residual error jointly by controlled-MCMC stochastic approximation. The fitted conditional model has an exact conventional FFEM representation when parameter margins are Normal and an exact conditional generative representation otherwise. An optional outer selection step ranks eta-margin families with independent full-posterior samples before a fresh fixed-model fit. In 400 simulated nonlinear mixed-effects datasets, structural-parameter bias remained within 0.47%. Relative to Gaussian FREM, mean differences in observed log likelihood were 4.30–9.41 under non-Gaussian data-generating distributions and -0.06 under a Gaussian data-generating distribution. Among subjects above the 95th percentile of a Gamma-distributed covariate, flexible margins reduced the log-scale RMSE of VPC quantiles by 54%. These results indicate that marginal-distribution assumptions can materially affect conditional prediction while Gaussian FREM remains available as a special case.

Keywords: nonlinear mixed-effects model; full random-effects model; Gaussian copula; stochastic approximation; marginal likelihood; covariates

INTRODUCTION

Covariate models describe how patient characteristics contribute to variability in individual model parameters and predictions. Full random-effects modelling (FREM) includes a prespecified set of covariates as additional random effects and estimates their joint covariance with the parameter random effects [1]. Covariate-effect coefficients and unexplained parameter variability can then be calculated for any subset of the included covariates from a single estimation. This avoids repeated estimation of separate reduced covariate models and allows missing covariates to be integrated over in the likelihood [2, 3].

Standard FREM assumes that the combined parameter and covariate random effects are multivariate Normal. For covariates observed with negligible error, this working model can estimate means and covariances accurately under some non-Normal covariate distributions [1]. Recent FREM investigations also show that covariate-model misspecification can affect inclusion and omission bias [4]. The Gaussian working model nevertheless specifies the entire population distribution, including marginal shape and conditional tail behaviour. These features are relevant when the fitted model is used for conditional prediction, simulation of virtual populations, integration over missing covariates, or extrapolation to sparsely represented covariate values.

A copula permits the marginal distributions to be specified separately from their dependence [5, 6]. We extend FREM by combining a Gaussian copula with prespecified continuous margins for parameter random effects and continuous or ordered categorical margins for covariates. The multivariate Normal FREM model is recovered when all margins are Normal. Estimation is based on the joint observed-data likelihood of the responses and recorded covariates, with individual parameters and missing covariates treated as unobserved data.

The objectives of this work were to define the flexible-margin FREM model, derive its observed-data likelihood and an exact FFEM translation, develop a likelihood-based stochastic-approximation estimator, develop

posterior-based eta-margin selection without changing families during estimation, and evaluate parameter recovery and conditional prediction in simulation studies.

METHODS

FREM model

For subject i at design point x_{ij} , write the nonlinear mixed-effects observation model as [7]

$$y_{ij} = f(\psi_i, x_{ij}) + g(\psi_i, \sigma, x_{ij})\epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, 1), \quad (1)$$

where ψ_i is the vector of individual model parameters and σ contains the residual-error parameters. On the transformed additive scale,

$$\phi_i = h(\psi_i) = C_i\mu + \eta_{\text{par},i}, \quad (2)$$

where C_i is the fixed-effect design matrix, μ is the vector of population parameters, and $\eta_{\text{par},i}$ is the parameter random effect. The lowercase symbol c_i denotes the FREM covariate vector and distinguishes it from the design matrix C_i . An error-free additive FREM covariate model is

$$c_i = \mu_c + \eta_{\text{cov},i}. \quad (3)$$

Here $\eta_{\text{par},i}$ represents total parameter variability (TPV) and $\eta_{\text{cov},i}$ is the individual covariate deviation. Standard FREM specifies

$$\begin{pmatrix} \eta_{\text{par},i} \\ \eta_{\text{cov},i} \end{pmatrix} \sim N[0, \Omega_{\text{FREM}}], \quad \Omega_{\text{FREM}} = \begin{pmatrix} \Omega_{\text{par}} & \Omega_{\text{par,cov}} \\ \Omega_{\text{cov,par}} & \Omega_{\text{cov}} \end{pmatrix}. \quad (4)$$

Equation (4) is the standard FREM specification [8, 1]. Let $\Omega_{\text{par,cov}} \in \mathbb{R}^{p \times q}$ and $\Omega_{\text{cov,par}} = \Omega_{\text{par,cov}}^\top$. The covariate-effect matrix and unexplained parameter variability (UPV) are

$$B = \Omega_{\text{par,cov}}\Omega_{\text{cov}}^{-1}, \quad \Omega'_{\text{par}} = \Omega_{\text{par}} - B\Omega_{\text{cov,par}}. \quad (5)$$

Consequently,

$$\eta_{\text{par},i} \mid c_i \sim N[B(c_i - \mu_c), \Omega'_{\text{par}}]. \quad (6)$$

Equations (4)–(6) give the standard Gaussian FREM model, its conditional covariate coefficients, and its UPV representation.

Flexible-margin FREM model

Let $X_i = (\eta_{\text{par},i}, c_i)$ contain the d population coordinates. For continuous coordinate j , write $F_j(\cdot; \alpha_j)$ and $f_j(\cdot; \alpha_j)$ for its CDF and density and define

$$\xi_j = \Phi^{-1}\{F_j(x_j; \alpha_j)\}.$$

Let $R(\rho)$ be a positive-definite Gaussian-score correlation matrix and let $\theta_p = (\alpha, \rho)$ collect the population-distribution parameters. The continuous population density is

$$p_{\theta_p}(x) = |R|^{-1/2} \exp\left[-\frac{1}{2}\xi^\top(R^{-1} - I)\xi\right] \prod_{j=1}^d f_j(x_j; \alpha_j). \quad (7)$$

We parameterize R by a fixed regular vine with Gaussian pair copulas; the vine structure is not estimated during the stochastic-approximation recursion. When all margins are Normal on their modelled scales, (7) reduces exactly to the error-free Gaussian FREM population model (2)–(4), with Ω_{FREM} obtained from the marginal standard deviations and R . This error-free formulation is the zero-covariate-residual limit of the conventional NONMEM/PsN formulation, which uses a small fixed residual variance for covariate responses [1, 3].

Let $\theta = (\mu, \alpha, \rho, \sigma)$ denote the unknown parameter vector. Here α contains the parameters of the marginal distributions of η_{par} and c , including μ_c ; ρ parameterizes R ; and σ contains the residual-error parameters. When all margins are Normal, the marginal scale parameters together with ρ are equivalently represented by $\text{vec}(\Omega_{\text{FREM}})$; the covariate locations μ_c remain separate.

The corresponding FREM effect summaries live naturally on the latent-score scale. Partition R into individual-parameter and covariate blocks. Then

$$B_\xi = R_{\text{par}, \text{cov}} R_{\text{cov}}^{-1}, \quad R'_{\text{par}} = R_{\text{par}} - B_\xi R_{\text{cov}, \text{par}}, \quad (7a)$$

so that $\xi_{\text{par}} \mid c$ is Normal with mean $B_\xi \xi_{\text{cov}}(c)$ and covariance R'_{par} . The fitted marginal quantile maps transform this conditional law back to the native parameter scale. Under all-Normal margins, let D_{par} and D_{cov} be diagonal matrices of marginal standard deviations. Then $B = D_{\text{par}} B_\xi D_{\text{cov}}^{-1}$ and $\Omega'_{\text{par}} = D_{\text{par}} R'_{\text{par}} D_{\text{par}}$, which recovers (5). Under non-Gaussian parameter margins the native-scale conditional mean need not be linear. In that case, B_ξ , R'_{par} and the marginal quantile functions jointly determine the conditional distribution.

Maximum likelihood estimation

For the Gaussian FREM model, the complete-data likelihood has the curved exponential-family representation used by standard SAEM,

$$\log p_\theta(Y, Z) = -\psi(\theta) + \langle S(Y, Z), \varphi(\theta) \rangle,$$

where $S(Y, Z)$ is a parameter-independent finite-dimensional sufficient statistic. Standard SAEM recursively averages $S(Y, Z)$ and performs the M-step from that finite vector [9, 10].

In (7), a marginal parameter α_j enters both $\log f_j(x_j; \alpha_j)$ and the transformed value $\xi_j = \Phi^{-1}\{F_j(x_j; \alpha_j)\}$ inside the copula quadratic form. Marginal and dependence parameters are therefore coupled through nonlinear functions of the latent data, and a parameter-independent finite statistic $S(Y, Z)$ is not generally available. The observed-data likelihood and the EM identity remain valid, but the usual finite-statistic SAEM recursion and its convergence theorem cannot simply be carried over; the curved-exponential condition is a recognized bottleneck for SAEM [11]. One possible workaround adds fixed-variance Gaussian latent variables to exponentialize the model, but then maximizes a perturbed likelihood whose approximation and numerical behaviour depend on that variance [12, 11]. Here the likelihood score of the original, unperturbed model is used instead, through the stochastic-gradient recursion proposed by Younes and presented as equation (74) by Delyon et al. [13, 9]. When all margins are Normal, the ordinary Gaussian FREM finite-statistic representation is recovered.

Observed-data likelihood

Let $O_i = (Y_i, c_{i, \text{obs}})$ and let Z_i collect individual parameters and missing covariates. For categorical covariates and continuous margins with parameter-dependent support, Z_i additionally contains the fixed-support unit-interval variables defined below. With a fixed mixed Lebesgue/counting measure λ , define the complete-data likelihood f_i and observed-data likelihood g_i by

$$g_i(\theta) = \int f_i(z_i; \theta) \lambda(dz_i), \quad \ell(\theta) = \sum_{i=1}^N \log g_i(\theta). \quad (8)$$

The dependence of f_i and g_i on O_i is implicit, as in Delyon et al.; Z_i denotes the missing or augmented data [9]. Consequently, the likelihood contribution is based on the joint density $p_\theta(Y_i, c_{i,\text{obs}})$. Individual effects and missing covariates are integrated out rather than replaced by point estimates. The analysis conditions on the recorded missingness pattern and assumes that the covariate-missingness mechanism is ignorable.

When all covariates are observed, its subject contribution has the exact factorization

$$\log p_\theta(Y_i, c_i) = \log p_{\theta_p}(c_i) + \log p_\theta(Y_i \mid c_i). \quad (8a)$$

Thus a difference in joint log likelihood may reflect differences in the fitted covariate density, the conditional response likelihood, or both. This factorization does not alter the estimation objective in (8).

When the continuous margins are proper and absolutely continuous, the required fixed-support representations exist, the categorical probability vectors are proper, R is positive definite, and the response density is proper, (7) defines a probability distribution and $g_i(\theta)$ in (8) is a valid observed-data likelihood contribution. This follows from a componentwise probability-integral transformation: continuous Jacobians transform (7) to a $N_d(0, R)$ density, categorical cells partition their latent Gaussian coordinates, and the fixed-support representations put moving boundaries on fixed unit intervals. Appendix A gives the detailed normalization argument.

Stochastic approximation of the likelihood score

Work in unconstrained internal coordinates ϑ , with native parameters $\theta = \tau(\vartheta)$. Log, logit and hyperspherical transformations enforce positive scales, probabilities and a positive-definite R . Write $\tilde{f}_i(z; \vartheta) = f_i\{z; \tau(\vartheta)\}$ and $\tilde{g}_i(\vartheta) = g_i\{\tau(\vartheta)\}$. At iteration k , controlled Markov kernels invariant for the current individual conditional law advance m chains per subject. The estimated complete score is

$$\hat{H}_{k+1} = \frac{1}{mN} \sum_{r=1}^m \sum_{i=1}^N \nabla_{\vartheta} \log f_i(Z_{i,k+1}^{(r)}; \tau(\vartheta_k)), \quad (9)$$

and the parameter update is

$$\vartheta_{k+1} = \text{Proj}_K \{\vartheta_k + \gamma_{k+1} A_k \hat{H}_{k+1}\}. \quad (10)$$

Adaptation of the positive diagonal metric and proposal scales is restricted to a finite initial sequence; these quantities are subsequently held fixed. The gain sequence is $\gamma_k = \gamma_0(k + k_0)^{-a}$ with $3/4 < a \leq 1$. Marginal families, category order, copula structure and all other discrete model choices are specified before recursion (10) begins.

Equation (10) is a controlled-Markov, internal-coordinate, projected, preconditioned and multiple-chain extension of the Younes stochastic-gradient recursion presented as Delyon et al. equation (74), $\theta_k = \theta_{k-1} + \gamma_k \partial_{\theta} \log f(Z_k; \theta_{k-1})$. The coordinate transformations and fixed positive-definite metric leave the stationary points unchanged. Because the implemented chains persist between iterations, their draws are Markovian and do not satisfy the conditional-independence assumption used in Delyon et al.'s direct analysis of equation (74). Appendix B replaces that martingale-difference argument by the Poisson-equation decomposition for controlled-Markov stochastic approximation, building on the SAEM-MCMC construction and its later extensions [14, 15].

Every stochastic move is either an exact conditional draw or a Metropolis-Hastings transition with the current individual conditional law as invariant distribution. At least one exact conditional-population independence proposal is used per iteration. With several categorical covariates, random-walk moves that would require a numerical rectangle probability are omitted; the exact independence kernel remains. An unadjusted approximate kernel would generally have a different invariant distribution and hence a different mean field [16].

Relation to the observed-data likelihood

Provided differentiation can pass through (8), Fisher's identity gives

$$E_{\vartheta}\{\nabla_{\vartheta} \log \tilde{f}_i(Z_i; \vartheta) \mid O_i\} = \nabla_{\vartheta} \log \tilde{g}_i(\vartheta). \quad (11)$$

For one subject, substitution of the conditional density $\tilde{f}_i(z; \vartheta)/\tilde{g}_i(\vartheta)$ into the expectation reduces its numerator to $\nabla_{\vartheta} \int \tilde{f}_i(z; \vartheta) \lambda(dz)$. The fixed-support representations ensure that λ is fixed. Appendix A states the required domination conditions and gives the full calculation [17].

Let $\bar{\ell} = N^{-1}\ell$. Thus the mean field after finite tuning is $A\nabla\bar{\ell}$. For symmetric positive-definite A ,

$$\frac{d}{dt} \bar{\ell}\{\vartheta(t)\} = \nabla\bar{\ell}(\vartheta)^{\top} A \nabla\bar{\ell}(\vartheta) \geq 0. \quad (12)$$

Convergence

Let $\mathcal{L} = \{\vartheta : \nabla\bar{\ell}(\vartheta) = 0\}$. Assume:

1. the model, augmentation and dominating measure are fixed, and $\bar{\ell}$ is continuously differentiable on a neighbourhood of the visited compact interior set;
2. differentiation under the integral is dominated, category probabilities and eigenvalues of R stay away from zero, and moving-support quantile and response paths have dominated derivatives;
3. the controlled kernels have unique invariant laws, uniform geometric drift and minorization on the visited compact set, and the kernel and score satisfy the local regularity and Poisson-equation bounds required for Markovian stochastic approximation;
4. $\gamma_k = \gamma_0(k + k_0)^{-a}$ satisfies the usual summability conditions and the stronger controlled-Markov requirement; the value $a = 0.80$ used in the experiments is valid for the stated Lipschitz and second-moment setting;
5. the metric is eventually fixed and positive definite, numerical score error is gain-weighted summable, and projections, invalid-step backtracking and no-move replacements cease after a finite random time; and
6. $-\bar{\ell}$ admits the required coercive Lyapunov extension and critical-value condition on the stabilized tail.

Then controlled-Markov stochastic-approximation results imply

$$d(\vartheta_k, \mathcal{L}) \longrightarrow 0 \quad \text{almost surely.} \quad (13)$$

The limit set lies in a connected component of $\mathcal{L}$. If it is contained in a neighbourhood with one isolated attracting stationary point, the raw iterates converge to that point. This result establishes convergence to the stationary set, not convergence to a global maximum. It is distinct from Delyon et al. Theorems 5 and 8, which concern finite-statistic SAEM. Appendix B maps the assumptions to controlled-Markov results [18, 15].

Posterior-based eta-margin selection

Eta-margin families are discrete model choices and are not changed during recursion (10). They may instead be screened before a fresh fixed-model fit by reusing the complete individual posterior distribution from a fitted Gaussian incumbent. Let f_{0i} and g_{0i} denote the incumbent complete- and observed-data likelihoods, and let $\pi_{0i}(z) = f_{0i}(z)/g_{0i}$ be its individual conditional distribution. For a candidate model m with the same observation model, population locations, covariate margins and Gaussian dependence during screening,

$$\frac{g_{mi}}{g_{0i}} = E_{\pi_{0i}} \left\{ \frac{f_{mi}(Z_i)}{f_{0i}(Z_i)} \right\} = E_{\pi_{0i}} \left\{ \frac{p_m(\eta_i, c_i)}{p_0(\eta_i, c_i)} \right\}. \quad (14)$$

Thus full posterior draws, unlike EBEs, retain individual uncertainty and give an observed-likelihood comparison for the screened population change. The implementation generates two posterior pools with independent random-number streams. Candidate margin parameters are optimized on the first pool. The second pool gives the validation criterion

$$2 \sum_{i=1}^N \log \left[\frac{1}{M} \sum_{r=1}^M \frac{p_m(\eta_i^{(r)}, c_i)}{p_0(\eta_i^{(r)}, c_i)} \right] - (k_m - k_0) \log N, \quad (15)$$

which is the candidate's validation BIC advantage relative to the incumbent. Under draws from π_{0i} , the inner importance average is unbiased for g_{mi}/g_{0i} , but its logarithm is downward-biased by Jensen's inequality. The incumbent ratio equals one exactly and has no corresponding Monte Carlo bias, so the finite-pool criterion is conservative toward retaining the incumbent; batch-means Monte Carlo standard errors describe sampling variability but do not remove this bias. The current candidate set comprises centred Normal, Student and Laplace eta margins. All combinations up to a prespecified cap are evaluated while the incumbent Gaussian dependence and non-eta margins are held fixed. Posterior-chain ESS and acceptance, subject-level importance ESS, batch-means Monte Carlo standard error and separation of the leading criterion from its runner-up are reported. An unresolved screen is not accepted automatically; posterior sampling must be increased or the scientifically prespecified candidate set reduced. In particular, a Student candidate screened from a Normal incumbent can have unbounded importance-weight variance because its likelihood ratio grows in the tails. The subject-level importance ESS is therefore an operative overlap guard, although a finite empirical ESS cannot prove existence of a second moment.

The selected family combination is subsequently frozen and passed to a fresh score stochastic-approximation fit, which re-estimates population locations, margin parameters, Gaussian dependence and residual error. Consequently, the convergence result in (13) applies to the fresh fixed-model fit, not to the preceding discrete selection operation. Final model assessment may additionally compare independently evaluated observed likelihoods after refitting the leading candidates.

Conditional representation and FFEM translation

For a selected covariate subset S , let z_{iS} contain the corresponding Gaussian scores and calculate $B_{\xi,S}$ and $R'_{\text{par},S}$ from the relevant blocks of R . If the parameter random-effect margins are Normal, the fitted conditional distribution has the exact FFEM representation

$$\phi_i = C_i \mu + \beta_{z,S} z_{iS} + \eta'_{iS}, \quad \beta_{z,S} = D_{\text{par}} B_{\xi,S}, \quad \eta'_{iS} \sim N(0, \Omega'_{\text{par},S}),$$

where $\Omega'_{\text{par},S} = D_{\text{par}} R'_{\text{par},S} D_{\text{par}}$. The covariates are fixed inputs in this FFEM; their fitted CDFs are used only to calculate z_{iS} . If the selected covariate margins are also Normal, the transformation reduces to centring and scaling, and the usual raw-scale B_S and $\Omega'_{\text{par},S}$ of standard FREM are recovered [1].

For non-Normal parameter random effects, a conventional additive-Normal FFEM is not generally equivalent. Let Q_η apply the fitted parameter-margin quantile functions componentwise and choose L_S such that $L_S L_S^\top = R'_{\text{par},S}$. The exact conditional representation is

$$\phi_i = C_i \mu + Q_\eta[\Phi\{B_{\xi,S} z_{iS} + L_S \varepsilon_i\}], \quad \varepsilon_i \sim N(0, I).$$

The covariates remain fixed inputs, but the latent Normal random effects are transformed to retain the fitted non-Normal parameter margins. We use *distributional FFEM* as shorthand for this exact conditional generative representation, not for a conventional additive-Normal FFEM. Ordered categorical covariates similarly require their latent-interval mixture rather than one deterministic transformed value. The implementation exports the linear coefficient and UPV matrices when the conventional FFEM is exact; otherwise the fitted conditional model is represented by exact conditional quantiles and simulation, without a Normal approximation. The translated model represents $p(y | c_S)$ and therefore omits the covariate-density contribution to the joint FREM likelihood. Appendix A derives the Gaussian nested case and both FFEM representations.

Categorical covariates and parameter-dependent support

For an ordered categorical covariate j with probabilities $\pi_{j1}, \dots, \pi_{jK}$, category k corresponds to the latent Gaussian interval between cumulative-probability thresholds. Several observed categories therefore contribute a joint multivariate-Normal rectangle mass. The estimator avoids differentiating this rectangle by introducing $V_{ij} \in (0, 1)$ and

$$U_{ij} = b_{j,k-1} + V_{ij}\pi_{jk}, \quad b_{jk} = \sum_{r \leq k} \pi_{jr}. \quad (16)$$

Integrating V_{ij} recovers the category mass. This construction supports any finite number of dependent binary, ordinal, or explicitly ordered nominal covariates. A permutation-invariant nominal model instead requires a multinomial latent-utility construction.

A regular continuous margin whose support depends on α_j uses the analogous fixed-percentile representation

$$W_{ij} \in (0, 1), \quad X_{ij} = F_j^{-1}(W_{ij}; \alpha_j). \quad (17)$$

For a parameter coordinate with parameter-dependent support, the complete-data score differentiates $\phi_{ij} = (C_i\mu)_j + F_j^{-1}(W_{ij}; \alpha_j)$ through $\psi_i = h^{-1}(\phi_i)$ and the structural response. Missing covariates with parameter-dependent support use the same representation. Observed native values must remain in the interior of their support on the visited parameter set; nonregular endpoint MLEs are outside the stationary- interior theory.

Experiments

The simulation study used four covariate-margin scenarios: Normal/Normal, lognormal/Normal, Gamma/Normal, and Weibull/lognormal. Each contained 100 independently generated datasets with $N = 100$, seven intravenous one-compartment observations per subject, $V = 10$, $CL = 3$, and proportional residual standard deviation 0.12. The four-dimensional Gaussian copula joined two individual parameters and two covariates. Flexible families were prespecified to isolate estimator behaviour from family selection. In the coordinate order $(\eta_V, \eta_{CL}, \text{COV1}, \text{COV2})$, the six generating Gaussian-score correlations were 0.25, 0.50, 0.10, 0.25, 0.55 and 0.20. COV1 was $N(70, 14^2)$, lognormal with meanlog $\log(70)$ and sdlog 0.24, Gamma with shape 7.5 and scale 12, or Weibull with shape 2.3 and scale 78. COV2 was $N(90, 25^2)$ in the first three scenarios and lognormal with meanlog $\log(90)$ and sdlog 0.28 in the fourth. Each fit used 1,500 score iterations, one persistent chain per subject, and two moves from each of the three invariant kernels per iteration. The gain was $0.2(k + 30)^{-0.80}$, with score multiplier 0.025, $h_0 = 10^{-4}$ and 50 learning iterations. A Gaussian FREM fit supplied only a finite initializer; all score-recursion state was restarted. For each dataset, the final score-SA and Gaussian-FREM fits were reevaluated by importance sampling with 5,000 draws and independent prespecified random-number seeds. A fit was considered successful only if it completed, adaptation ended as specified, and no projection, invalid-step backtracking, no-move replacement or one-sided numerical difference occurred after adaptation.

Because Gamma shape and scale can compensate while retaining the marginal mean, a separate direct-MLE experiment used 10,000 independently generated Gamma samples at each of $N = 50, 100, 300, 1,000$ and 3,000. Full-model Gamma estimates were also compared with direct Gamma MLEs on the same $N = 100$ and $N = 300$ datasets.

An additional example was used to evaluate prediction conditional on an extreme covariate value. One dataset with $N = 300$ was generated from the same one-compartment model. C-reactive protein (CRP) followed a Gamma distribution with shape 2 and scale 20, albumin followed $N(40, 5^2)$, and the Gaussian-score correlation between CRP and the clearance random effect was 0.55. Gaussian and flexible-margin FREM were fitted with 1,500 iterations. An independent validation population was then sampled above the 95th percentile of CRP. Only CRP was supplied for prediction; albumin was integrated over its fitted conditional distribution. Generating and fitted VPC quantiles were calculated from 150,000 virtual subjects using common random numbers. Empirical validation quantiles were calculated independently from 1,500 subjects. Predictive error was the RMSE of log-transformed 10th, 50th and 90th percentiles over seven time points.

The posterior selection identity was evaluated separately in an additive latent model, $Y_i = \eta_i + \epsilon_i$, where the direct observed likelihood is available analytically for Normal and Laplace eta margins. Twelve datasets per generating family were simulated with $N = 100$, eta standard deviation 0.45 and residual standard deviation 0.25. A fitted Gaussian incumbent supplied 2,000 posterior draws per subject for each of two independently generated pools. Candidate scale parameters were fitted on the first pool and family rankings were evaluated on the second. A weak-information setting used residual standard deviation 0.60; discordant close comparisons were repeated with 20,000 draws per subject.

RESULTS

Re-estimation study

Table 1: Flexible-margin FREM parameter recovery and difference in estimated joint observed-data log likelihood relative to Gaussian FREM. Bias is relative mean bias across 100 datasets; $\Delta\ell$ is flexible-margin minus Gaussian FREM.

Scenario	V bias (%)	CL bias (%)	Residual bias (%)	$\Delta\ell$
Normal/Normal	0.29	-0.03	0.03	-0.06
lognormal/Normal	-0.03	-0.46	-0.15	4.30
Gamma/Normal	0.31	0.18	-0.41	4.60
Weibull/lognormal	-0.26	0.01	0.05	9.41

Across scenarios, flexible-margin RMSE was 0.226–0.249 for V , 0.073–0.091 for CL , and 0.0033–0.0040 for residual standard deviation. The corresponding Gaussian-FREM results were nearly identical: the largest difference between methods in mean relative bias was 0.04 percentage points.

Estimates of the marginal-distribution parameters were close to their generating values. Mean lognormal sdlog estimates were 0.239 and 0.277 for generating values 0.240 and 0.280. Mean Weibull shape and scale estimates were 2.309 and 78.11 for generating values 2.3 and 78. Mean Gamma shape and scale estimates were 7.944 and 11.556 for generating values 7.5 and 12. On the same $N = 100$ datasets, direct Gamma MLEs were 7.941 and 11.562; mean absolute full-model-direct-MLE differences were 0.0065 and 0.0101. Thus the full model reproduced the dataset-specific Gamma MLE. Across the six Gaussian-score correlations, the maximum absolute scenario-specific mean bias was 0.022 and RMSE ranged from 0.066 to 0.110.

The estimated joint observed-data log likelihood was larger for the flexible-margin model in 285 of 300 datasets generated with a non-Gaussian margin: 90 lognormal/Normal, 95 Gamma/Normal and all 100 Weibull/lognormal datasets. Under the Gaussian generating model, the mean difference was -0.06 and exactly 50 of 100 differences were positive, as expected for two numerical estimators of the same model.

The covariate-density component accounted for mean log-likelihood differences of 4.01, 4.15 and 8.58 in the lognormal/Normal, Gamma/Normal and Weibull/lognormal scenarios. The corresponding mean differences in the conditional response likelihood were 0.29, 0.45 and 0.83. Thus most of the joint-likelihood gain arose from the specified covariate margins; the separate predictive evaluation below addresses the effect on conditional prediction.

For each dataset, likelihoods were evaluated by independent importance-sampling runs. The quadrature sum of the two reported Monte Carlo standard errors averaged 0.29. The compared covariate margins had equal numbers of parameters, but the models were non-nested; the reported differences are descriptive rather than a formal model comparison. Equation (8a) further shows that a difference in the joint log likelihood does not identify the corresponding difference in $p(Y | c)$.

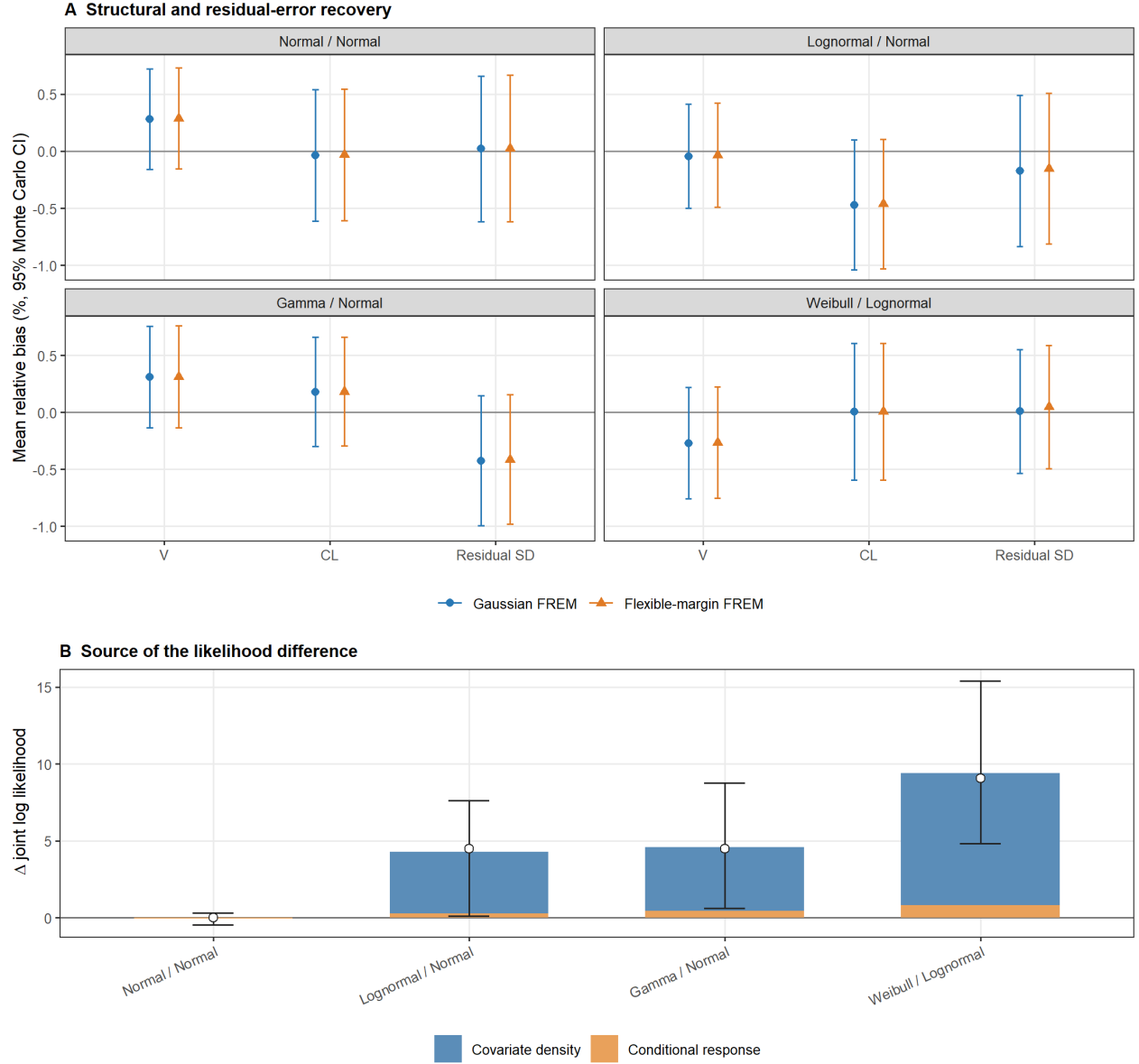


Figure 1: Re-estimation results from 100 independently generated datasets per scenario. (A) Mean relative bias and 95% Monte Carlo confidence intervals for the structural and residual-error parameters. (B) Mean decomposition of the difference in joint observed-data log likelihood into covariate-density and conditional-response components. Open circles and whiskers show the median and 10th–90th percentiles of the total likelihood difference.

Predictive evaluation

Both methods recovered the structural and residual-error parameters in the additional example: estimates of V , CL and residual standard deviation were 10.16, 3.01 and 0.120 for Gaussian FEM and 10.16, 3.00 and 0.120 for flexible-margin FEM. The paired estimated joint log likelihood was -4091.6 and -4041.7, respectively. Because the models are non-nested and the joint likelihood includes the covariate density, this difference is descriptive.

The fitted conditional clearance relationships are shown in Figure 2. The Gaussian-FREM relationship is the exact FFEM transformation implied by its fitted Normal CRP margin and covariance matrix. It followed the centre of the simulated data but increased too rapidly in the upper CRP tail. The flexible- margin FEM

transformation remained closer to the generating relationship.

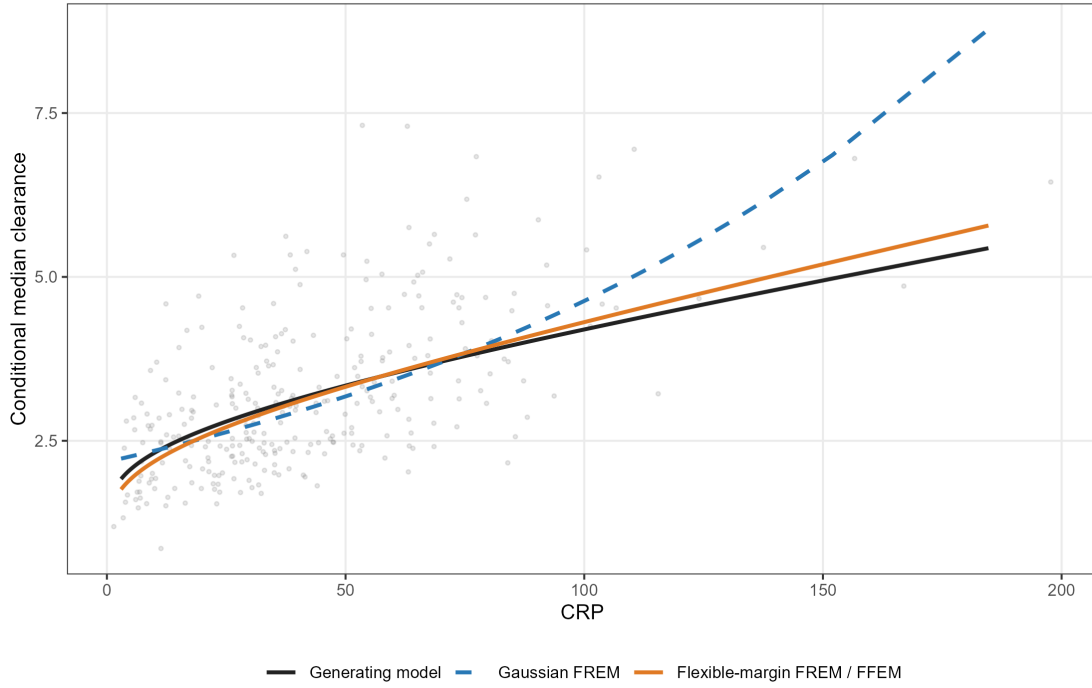


Figure 2: Conditional median clearance as a function of CRP in the simulated $N = 300$ dataset. Grey points are the true simulated individual clearance values, which are latent rather than directly observed in an NLME analysis. Lines show the generating conditional median and the relationships obtained from Gaussian FREM and flexible-margin FREM after exact FFEM translation. Only CRP is conditioned on; albumin is marginalized from the joint distribution.

In the independent high-CRP population, Gaussian FREM underestimated the lower response percentile and overestimated the upper response percentile at later times. The flexible-margin predictions were closer to the generating percentiles throughout the profile. Log-scale VPC RMSE decreased from 0.266 to 0.123, a reduction of 53.9%. Albumin was not supplied for prediction and was integrated out under both fitted joint distributions.

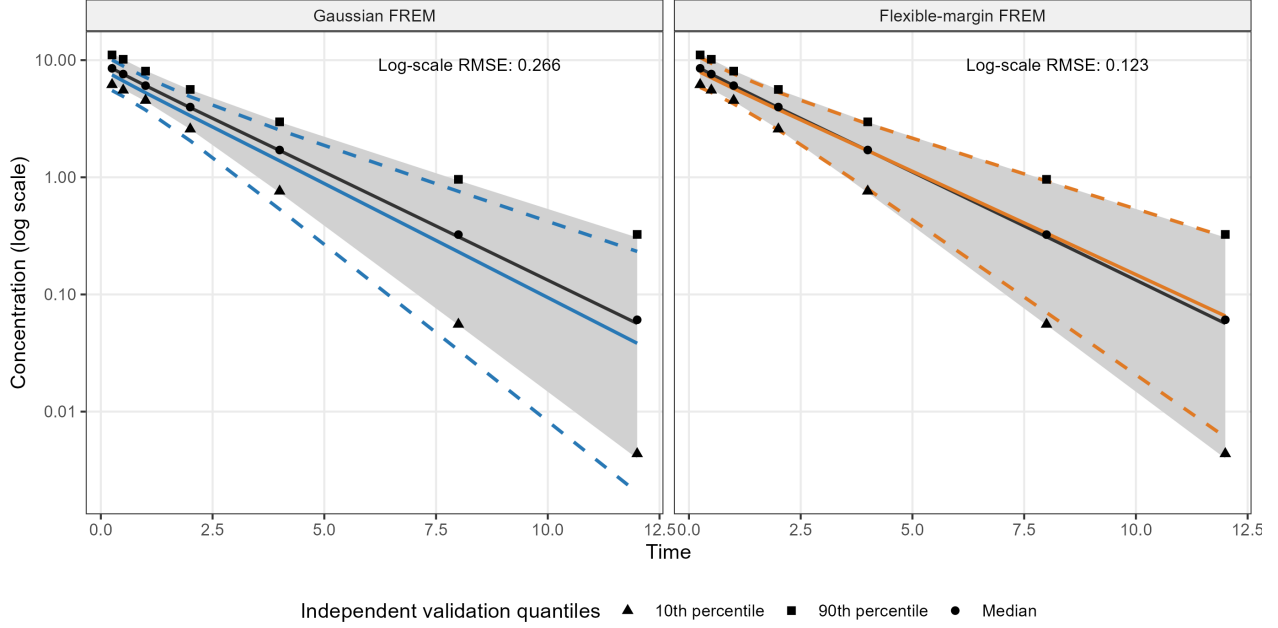


Figure 3: Conditional VPC in an independent population above the 95th CRP percentile. Grey shading and the black line show the generating 10th–90th percentile interval and median. Coloured solid and dashed lines show fitted median and 10th/90th percentiles; black points show empirical quantiles in the independent validation sample. Only CRP was available for prediction; albumin was integrated out. Flexible-margin FEM reduced log-scale VPC RMSE from 0.266 to 0.123.

Numerical verification

Independent numerical derivatives agreed with analytical complete-data scores to 1.7×10^{-10} for Gamma and Weibull examples and 6.5×10^{-8} for a lognormal example. Monte Carlo evaluations of Fisher’s identity included population parameters μ , marginal parameters, dependence parameters and residual error. With two categorical covariates, the maximum absolute discrepancy was 0.0062, compared with a maximum Monte Carlo standard error of 0.0113. For a parameter-dependent support example, the maximum absolute discrepancy was 2.2×10^{-10} . Additional numerical tests included multiple categorical covariates, centred Uniform and Gamma parameter random effects, and missing Uniform covariates.

In the direct Gamma-MLE experiment, mean shape bias decreased from 2.83% at $N = 100$ to 0.14% at $N = 3,000$, and mean scale bias decreased from -0.85% to -0.07%; bias in their product remained within 0.05%. Mean absolute full-model–direct-MLE differences in shape and scale were 0.0043 and 0.0075 in the $N = 300$ datasets. These results show that the shape–scale displacement in the $N = 100$ re-estimation study reflected finite-sample Gamma-MLE behaviour and simulation-replicate variation rather than score-recursion error.

For Normal parameter and covariate margins, exported raw-scale coefficients and UPV covariance agreed with the direct FEM block formulas to less than 10^{-12} . With a Gamma covariate and Normal parameter margins, 100,000 draws from the translated FFEM reproduced its analytical conditional means within 0.003 and covariance elements within 0.002. With a centred Gamma parameter margin, analytical conditional quantiles agreed with 150,000 exact conditional draws within 0.006. In the fitted CRP example, direct copula-FREM and translated FFEM conditional densities were identical to numerical precision, and the translated and direct concentration predictions were identical for every tested draw.

In the informative posterior-selection experiment, the posterior criterion and direct observed likelihood selected the same family in all 24 datasets. Both selected Normal in 12 of 12 Normal datasets. Both selected Laplace in 11 of 12 Laplace datasets; the remaining finite dataset was ranked as Normal by the direct likelihood as well.

The RMSE of the posterior estimate of the direct candidate log likelihood was below 0.001 for Normal candidates and 0.071 for Laplace candidates, with the larger error reflected by their reported Monte Carlo standard errors.

With weaker individual information and 2,000 posterior draws, the posterior and direct rankings agreed in 22 of 24 datasets. The two discordant datasets had direct Laplace-versus-Normal log-likelihood advantages of 0.164 and 0.114 and low tail-overlap diagnostics. Repeating those comparisons with 20,000 draws recovered the direct Laplace ranking in both cases. These results support adaptive posterior sampling and explicit overlap gates rather than a fixed draw count for automatic selection.

DISCUSSION

The proposed extension replaces the multivariate Normal population distribution of FREM by a Gaussian-copula distribution with prespecified margins; the structural and residual-error models are unchanged. When all margins are Normal, the two population distributions coincide. In the non-Gaussian simulation scenarios, the flexible-margin model gave larger estimated joint log likelihoods while structural-parameter biases remained small for both approaches. The principal difference was observed in prediction conditional on covariate-tail values. A misspecified marginal CDF changes the Gaussian score assigned to a covariate value and therefore changes the fitted conditional distribution $p(\eta_{\text{par}} | c)$, even when the first two moments are estimated accurately.

The conditional representation provides an explicit covariate model for a chosen subset. With Normal parameter margins it is a conventional FFEM using fixed transformed covariates. With non-Normal parameter margins it retains the fitted distribution through a latent-Normal quantile transformation. Direct copula-FREM and translated-model predictions are identical when the fitted population parameters are held fixed. Re-estimation of the translated model instead targets $p(y | c_S)$ and does not include the covariate-density term from the joint FREM likelihood.

The lack of a general finite sufficient statistic changes the stochastic-approximation step but not the likelihood being estimated. Population parameters were updated with the controlled-Markov extension of the Younes stochastic-gradient recursion presented by Delyon et al. Fisher’s identity shows that its mean field equals the gradient of the specified observed-data log likelihood. The convergence result is conditional on the stated stability, drift, regularity and critical-value assumptions. The reported numerical diagnostics establish that the finite-run stabilization criteria were satisfied, but do not establish these asymptotic assumptions.

The posterior eta-margin screen uses the same observed-likelihood principle without treating conditional modes as population observations. Its independent validation pool limits optimization bias, while ESS and Monte Carlo error make weak overlap visible. The Gaussian incumbent has full real support for the currently screened families, but heavier-tailed candidates can still produce high-variance likelihood ratios. Such candidates require additional draws or a defensive proposal; they should not be discarded merely because a small posterior pool is unstable. Family selection remains outside the score recursion, and the selected model is always estimated afresh.

Model-based copulas have previously been used in pharmacometrics to generate virtual populations from covariate distributions [19]. The present extension concerns likelihood-based estimation of the joint parameter-covariate distribution within a nonlinear mixed-effects model.

The simulation is exploratory: one PK model, 100 datasets per scenario, prespecified families and a Gaussian finite initializer were used. Coverage, observed-information standard errors, cold-start sensitivity, broader posterior-selection candidate libraries and powered categorical or moving-support studies remain future work. General non-Gaussian R-vines require additional score calculations and are outside the fixed Gaussian-copula scope. The continuous margins considered here must be proper, differentiable and identifiable with finite score moments on the stabilized compact set; parameter random-effect margins are centred and require finite variance. These conditions include many non-Gaussian families but do not define a distribution-free estimator. Nonregular support-boundary MLEs and permutation-invariant nominal variables also require separate theory. The current implementation also requires every estimated structural parameter to belong to the population-location block; estimated fixed-only structural parameters are outside its scope.

CONCLUSION

Flexible-margin Gaussian-copula FREM separates the specification of marginal distributions from the specification of dependence and reduces to Gaussian FREM when all margins are Normal. Fisher’s identity links the mean field of the controlled-MCMC stochastic-gradient algorithm to the observed-data likelihood score. In the simulations considered here, structural-parameter recovery was similar under both approaches, whereas flexible margins provided larger estimated joint log likelihoods and more accurate prediction conditional on extreme covariate values under non-Gaussian data-generating distributions. The fitted conditional model can be exported without loss to a conventional FFEM when parameter margins are Normal and to the exact conditional generative representation otherwise. Eta-margin families can be screened from full incumbent-posterior draws, with independent validation and overlap diagnostics, before beginning the fresh fixed-model fit to which the convergence result applies.

DECLARATIONS

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Likelihood normalization, Fisher identity and FFEM translation

Continuous normalization

For fixed θ , the map $x_j \mapsto \xi_j = \Phi^{-1}\{F_j(x_j)\}$ has Jacobian $d\xi_j/dx_j = f_j(x_j)/\phi(\xi_j)$. Substitution in (7) therefore gives

$$p_{\theta_p}(x) dx = \phi_R(\xi) d\xi.$$

Integration over $\mathbb{R}^d$ is one. This proves normalization for any proper absolutely continuous margins; Gaussian marginal shape is not required.

Several categorical covariates

For observed category vector $k_{\mathcal{D}}$, let $u_j = b_{j,k_j-1} + v_j \pi_{j,k_j}$ for $j \in \mathcal{D}$. The augmented mixed density is

$$p_{\theta}^{\text{aug}}(x_{\mathcal{C}}, k_{\mathcal{D}}, v_{\mathcal{D}}) = c_R(u) \prod_{j \in \mathcal{C}} f_j(x_j) \prod_{j \in \mathcal{D}} \pi_{j,k_j}. \quad (\text{A1})$$

Since $du_j = \pi_{j,k_j} dv_j$, integrating $v_{\mathcal{D}} \in (0,1)^{|\mathcal{D}|}$ gives the Gaussian-copula rectangle mass for $k_{\mathcal{D}}$. Summing category vectors tiles the latent unit cube, so the masses sum to one. Missing categories add counting-measure summation and remain normalized.

Moving continuous support

For $X = F_{\alpha}^{-1}(W)$ with $W \in (0,1)$, $dW = f_{\alpha}(X)dX$. Thus a latent moving-support coordinate contributes the fixed reference dW and its marginal density factor disappears after change of variables. Dependence still enters through $\xi = \Phi^{-1}(W)$. If the coordinate affects the response, $p_{\theta}\{Y \mid h^{-1}(C\mu + F_{\alpha}^{-1}(W))\}$ is differentiated through both μ and F_{α}^{-1} . The moving-boundary derivative is therefore represented by the pathwise quantile derivative rather than discarded.

Observed likelihood and Fisher identity

Multiplying the normalized population density by the conditional response density and integrating over all unobserved coordinates gives $g_i(\theta)$. Under an integrable derivative envelope,

$$\begin{aligned} E_{\vartheta}[\nabla \log \tilde{f}_i(Z_i; \vartheta) \mid O_i] &= \int \frac{\nabla \tilde{f}_i(z; \vartheta)}{\tilde{f}_i(z; \vartheta)} \frac{\tilde{f}_i(z; \vartheta)}{\tilde{g}_i(\vartheta)} \lambda(dz) \\ &= \frac{\nabla \int \tilde{f}_i(z; \vartheta) \lambda(dz)}{\tilde{g}_i(\vartheta)} = \nabla \log \tilde{g}_i(\vartheta). \end{aligned}$$

The chain rule gives $\nabla_{\vartheta} \bar{\ell} = J_{\tau}(\vartheta)^{\top} \nabla_{\theta} \bar{\ell}$. A nonsingular J_{τ} makes native- and internal- coordinate stationary points equivalent.

Gaussian nested model and FFEM translation

When every margin is Normal, write $X = \mu + D\xi$, where $\xi \sim N_d(0, R)$ and D is the diagonal matrix of marginal standard deviations. It follows directly that

$$X \sim N_d(\mu, DRD). \quad (\text{A2})$$

Thus the all-Normal Gaussian-copula model is the standard multivariate-Normal FREM model with $\Omega_{\text{FREM}} = DRD$; it is not an approximation to that model.

Let P index the parameter random effects and let S be any selected subset of continuous covariates. Marginalizing covariates outside S leaves the Gaussian subvector (ξ_P, ξ_S) with the corresponding submatrix of R . The multivariate-Normal conditioning formula gives

$$\xi_P \mid \xi_S = z_S = B_{\xi, S} z_S + L_S \varepsilon, \quad B_{\xi, S} = R_{P, S} R_{S, S}^{-1}, \quad L_S L_S^{\top} = R_{P, P} - B_{\xi, S} R_{S, P}, \quad (\text{A3})$$

where $\varepsilon \sim N(0, I)$. If the parameter margins are Normal, $\eta_P = D_P \xi_P$; substitution in (A3) yields the coefficient matrix $D_P B_{\xi, S}$ and UPV covariance $D_P L_S L_S^{\top} D_P$ stated in the conventional FFEM representation. If the selected covariate margins are also Normal, replacing z_S by $D_S^{-1}(c_S - \mu_S)$ gives the usual raw-scale FREM coefficient matrix and Schur-complement UPV covariance.

For continuous non-Normal parameter margins, let Q_{η} apply their quantile functions componentwise. Since $\eta_P = Q_{\eta}\{\Phi(\xi_P)\}$, (A3) gives

$$\eta_P \mid c_S = Q_{\eta}[\Phi\{B_{\xi, S} z_S(c_S) + L_S \varepsilon\}]. \quad (\text{A4})$$

Equation (A4) is the exact non-Normal conditional representation in the main text and has the same conditional law as the fitted FREM. For an ordered categorical covariate, conditioning is instead on its latent Gaussian interval; integrating (A3) over the corresponding truncated-Normal score distribution gives the stated interval mixture. Finally, direct FREM and translated-FFEM predictions integrate the same response kernel $p(Y \mid \eta_P)$ against this same conditional distribution. Their conditional predictive distributions are therefore identical when the population parameters are held fixed.

Posterior likelihood-ratio identity

Let f_{0i} and f_{mi} be incumbent and candidate complete-data densities with respect to the same fixed measure λ , and assume $f_{mi} = 0$ wherever the ratio is otherwise undefined. Since $\pi_{0i} = f_{0i}/g_{0i}$,

$$\begin{aligned} E_{\pi_{0i}}\left(\frac{f_{mi}}{f_{0i}}\right) &= \int \frac{f_{mi}(z)}{f_{0i}(z)} \frac{f_{0i}(z)}{g_{0i}} \lambda(dz) \\ &= \frac{1}{g_{0i}} \int f_{mi}(z) \lambda(dz) = \frac{g_{mi}}{g_{0i}}. \end{aligned} \quad (\text{A5})$$

When only the eta population distribution changes, the common response density cancels from f_{mi}/f_{oi} , giving (14). The Gaussian incumbent is positive on the real eta space used by the current Normal, Student and Laplace candidates. Equation (A5) establishes the likelihood identity, but not finite- sample stability of its importance estimator. Finite second moments, adequate tail overlap and sufficient MCMC mixing remain numerical requirements. In particular, when the incumbent eta margin is Normal and a candidate is Student, the likelihood ratio can have an infinite second moment under the incumbent posterior. The reported subject-level importance ESS detects severe realized weight concentration but cannot certify the missing moment. Moreover, although the importance average is unbiased for the likelihood ratio, applying the logarithm produces the conservative Jensen bias described after (15). Optimizing candidate parameters and validating their ranking on independently generated posterior pools separates fitting noise from the reported comparison. Once a family is selected, the posterior pools are discarded and a fresh fixed- model score fit is initialized.

Controlled kernel and convergence details

Exact conditional-population proposal

Let $q_{\vartheta,i}(\psi)$ denote the population density of ψ_i conditional on observed covariates, from which proposals are sampled exactly, and let $L_{\vartheta,i}(\psi) = p_{\theta}(Y_i | \psi, x_i)$. Then

$$\pi_{\vartheta,i}(\psi) = \frac{L_{\vartheta,i}(\psi)q_{\vartheta,i}(\psi)}{\int L_{\vartheta,i}(v)q_{\vartheta,i}(v)dv}.$$

Suppose uniformly on the compact parameter set that $L_{\vartheta,i}(\psi) \leq M_i$ and its normalizer is at least $m_i > 0$. The independence-Metropolis weight $w = \pi/q$ is bounded by M_i/m_i , and therefore

$$P_{\vartheta,i}^{\text{ind}}(\psi, A) \geq (m_i/M_i)\Pi_{\vartheta,i}(A). \quad (\text{B1})$$

Composition with any other invariant Metropolis kernels preserves the minorization. With finitely many subjects and chains, the product chain has a uniform Doeblin bound. Exact conditional augmentation after the eta move also preserves it: if $R_{\vartheta}(da | \psi')$ draws missing/categorical/fixed- reference variables, then

$$\tilde{P}_{\vartheta}\{(\psi, a), d\psi' da'\} = P_{\vartheta}(\psi, d\psi')R_{\vartheta}(da' | \psi') \geq \epsilon\Pi_{\vartheta}(d\psi', da'). \quad (\text{B2})$$

If a rejection sampler for a multivariate category rectangle reaches its finite iteration limit, the iteration is terminated without substituting an approximate draw.

Markovian stochastic-approximation decomposition

After finite adaptation, write the score noise as the conditional mean plus a martingale term, a Poisson-equation remainder and numerical error:

$$\hat{H}_{k+1} = \nabla \bar{\ell}(\vartheta_k) + e_{k+1} + r_{k+1}^{\text{P}} + r_{k+1}^{\text{num}}. \quad (\text{B3})$$

Here is the precise mapping to controlled-Markov stochastic approximation [20, 21, 14, 15]. On every visited compact parameter set, assume their H1–H2 with a drift function W : the partially marginalized score is uniformly W -bounded, each P_{ϑ} has unique invariant distribution Π_{ϑ} , the kernels are uniformly W -geometrically ergodic, and, for some $p \geq 2$,

$$\sup_{\vartheta} P_{\vartheta} W^p \leq \rho W^p + b, \quad \rho < 1. \quad (\text{B4})$$

The common minorization in (B1)–(B2), together with Lipschitz dependence of the kernel after finite adaptation, supplies H3 with exponent $v = 1$. H4 is the following integrated local-oscillation condition for the actual fixed-preconditioner increment $\mathcal{H}_{\vartheta}$: for every compact K and $\delta > 0$,

$$\sup_{\vartheta \in K} \int \Pi_{\vartheta}(dx) \sup_{\substack{\vartheta' \in K \\ \|\vartheta' - \vartheta\| \leq \delta}} \|\mathcal{H}_{\vartheta'}(x) - \mathcal{H}_{\vartheta}(x)\| \leq C_K \delta^{\alpha}, \quad \alpha = 1. \quad (\text{B5})$$

Smoothness of the mean field alone does not imply (B5); it is a stated regularity condition. Under H1–H4, the associated Poisson-equation solutions have the bounds used by Fort et al. Proposition 4.11. Their H6 requires a power-law gain exponent satisfying

$$\max\left\{\frac{1}{p}, \frac{1 + (\alpha \wedge v)/p}{1 + (\alpha \wedge v)}\right\} < a \leq 1. \quad (\text{B6})$$

For $p \geq 2$ and $\alpha = v = 1$, the implemented $a = 0.80 > 0.75$ satisfies (B6), not merely the usual Robbins–Monro sums.

After safeguards become inactive, apply Fort et al. Proposition 4.6 to the stopped tail recursion with mean field $h = A\nabla\bar{\ell}$. Its C-i follows from differentiability, C-ii from compact-tail stability, and C-iii from the gain. For H5 and C-v(A), assume a coercive continuously differentiable extension w that equals $-\bar{\ell} + C$ near the visited compact tail, has compact level sets and compact zero-drift set

$$\mathcal{L}_w = \{\vartheta : \langle \nabla w(\vartheta), h(\vartheta) \rangle = 0\},$$

such that $w(\mathcal{L}_w)$ has empty interior and $\mathcal{L}_w \cap K_{\text{tail}} = \mathcal{L} \cap K_{\text{tail}}$. This global extension and its critical-value property are assumptions, not automatic consequences of local likelihood smoothness.

It remains to establish Proposition 4.6 C-iv. Apply Fort et al. Proposition 4.11 and Corollary 4.12 to the process stopped on exit from the compact tail; eventual safeguard inactivity makes the exit time infinite from some finite random time onward. For its weighted controlled-Markov perturbation ξ_j , write

$$A_k = \sup_{l \geq k} \left\| \sum_{j=k}^l \gamma_j \xi_j \right\|.$$

Overlapping partial sums give $A_n \leq 2A_k$ whenever $n \geq k$. Hence, for every $\delta > 0$,

$$\{\limsup_n A_n \geq \delta\} \subseteq \bigcap_k \{A_k \geq \delta/2\}. \quad (\text{B7})$$

Corollary 4.12 gives $P(A_k \geq \delta/2) \rightarrow 0$, so the probability of the intersection is zero. A countable sequence of positive rational δ proves $A_k \rightarrow 0$ almost surely on the no-exit tail. Thus C-iv holds. Proposition 4.6 then yields $d(\vartheta_k, \mathcal{L}) \rightarrow 0$ almost surely. The finite prefix cannot change the limit set. This is a conditional stationary-set result for the Younes stochastic-gradient recursion presented as Delyon et al. equation (74), not an invocation of their finite-statistic SAEM Theorem 5 or 8.

Numerical score error

For centered differences, suppose

$$\mathcal{H}_{h_k}(\vartheta, u) = \mathcal{H}(\vartheta, u) + r_k(\vartheta, u), \quad \|r_k(\vartheta, u)\| \leq h_k^2 M(u),$$

with $\sup_k E\{M(Z_k)\} < \infty$. Taking $h_k = h_0\sqrt{\gamma_k}$, Tonelli’s theorem gives

$$E\left[\sum_k \gamma_k \|r_k\|\right] \leq C \sum_k \gamma_k h_k^2 = Ch_0^2 \sum_k \gamma_k^2 < \infty. \quad (\text{C1})$$

The nonnegative random series is therefore finite almost surely. This controls the complete random numerical perturbation, not merely its conditional mean. Post-freeze one-sided differences are rejected because they do not satisfy the same centered second-order error statement.

The reported endpoint averages internal-coordinate iterates after finite burn-in. If raw iterates converge to an isolated stationary point, their Cesaro average has the same limit. No asymptotic-efficiency claim for the average is made; Markov-chain trajectory averaging requires additional assumptions [22].

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